

20. Implement a basic information retrieval system using TF-IDF (Term Frequency-Inverse Document Frequency) for document ranking using python.

Code

```
from sklearn.feature_extraction.text import TfidfVectorizer
from sklearn.metrics.pairwise import cosine_similarity

documents = [
    "Python is a programming language",
    "Python is used for machine learning",
    "Machine learning uses artificial intelligence"
]

query = input("Enter search query: ")

vectorizer = TfidfVectorizer()

tfidf = vectorizer.fit_transform(documents + [query])

similarity = cosine_similarity(tfidf[-1], tfidf[:-1])

print("Document Rankings:")

for i, score in enumerate(similarity[0]):
    print("Document", i+1, ":", round(score, 3))
```

Input

machine learning

Output

Document Rankings:

Document 1 : 0.0

Document 2 : 0.632

Document 3 : 0.514

NATURAL LANGUAGE PROCESSING.ipynb

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```
[2] ✓ 12s
documents = [
    "Python is a programming language",
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print("Document Rankings:")
for i, score in enumerate(similarity[0]):
    print(f"Document", i+1, ":", round(score, 3))

... Enter search query: machine learning
Document Rankings:
Document 1 : 0.0
Document 2 : 0.448
Document 3 : 0.462
```

Variables Terminal

09:39 Python 3

30-07-2026